

MAE 4730
Lecture 39
11/30/2018

Today:

- Circles on circles
- steady precession of axisymmetric objects

Recall: The famous/simple solutions to 3D rigid objects dynamics

1) Fixed axis rotation  $\vec{\omega} = \omega \hat{i}$

key:

$$\dot{\vec{H}}/G = \underline{\underline{I}} \cdot \dot{\vec{\omega}} + \vec{\omega} \times \underline{\underline{I}} \cdot \vec{\omega}$$

$\begin{array}{c} \uparrow \omega \hat{i} \\ \boxed{\text{known}} \\ \text{rotates with object} \end{array} \quad \begin{array}{c} \uparrow \omega \hat{i} \end{array}$

2) Free motion

a) Euler equations on computer: $\vec{\omega} \& \underline{\underline{R}}$

b) Euler equations w/ body fixed coordinates

↑ $\rightarrow$ closed equations w/ just $[\vec{\omega}]_b$

c) b, linearized near prime axis rotation
 $\rightarrow$ stability results

3) Today: steady precession

object: axisymmetric

$$[\underline{\underline{I}}]_b = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_1 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

Motion: constant angular velocity relative to a precessing frame:

$$\vec{\omega}_{b/f} = \omega_s \hat{k}$$

$\omega_s = \text{constant}$

↑
precessing frame

· Constant rate precession $\vec{\omega}_{p/f} = \omega_p \hat{k}$

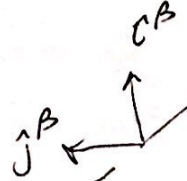
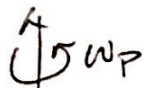
↳ constant

Dynamics Approach:

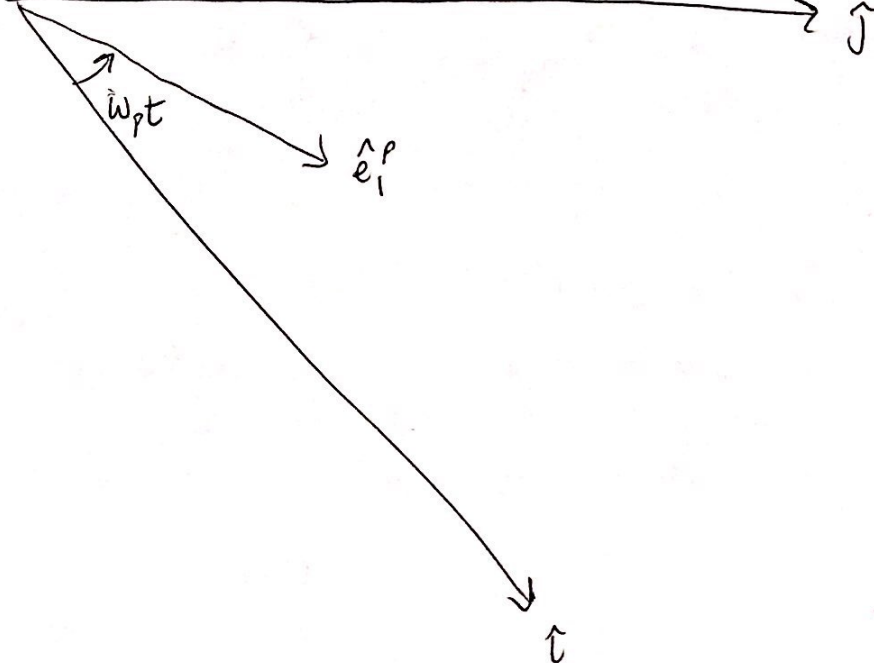
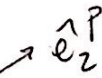
- 1) FBD
 - 2) Assume something about motion
 - 3) Characterized motion w/ motion variables
 - 4) AMB + LMB
 - 5) Solve for variables of interest
- maybe $\Rightarrow$ ODEs

$\Rightarrow$ Now: given $\omega_P, \omega_S, \hat{e}^P$, inertia properties
problem: calculate $\dot{\hat{H}}_{/G}$

$$\hat{e} = \hat{e}_3^P$$

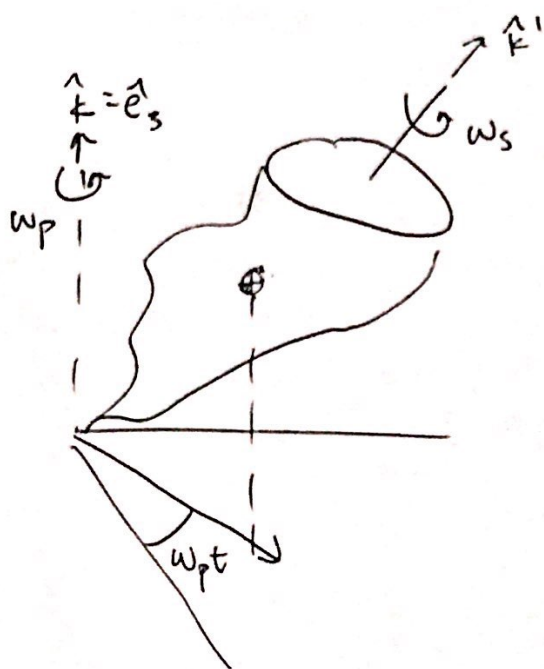


$\hat{e}^B = \text{constant in precessing frame}$
moves as rigid object w/ $\hat{e}_1^P + \hat{e}_2^P$



Assume: Axisymmetric

$$\underline{\underline{I}} = I_3 \hat{e}^B \hat{e}^B + I_1 \hat{e}^A \hat{e}^A + I_1 \hat{e}^B \hat{e}^B$$



$$\dot{\vec{H}}_{/G} = ?$$

$$\text{(given that)} \quad \vec{H}_{/G} = \vec{r}_{G/C} \times m \vec{a}_G + \vec{H}_{/G}$$

In principal want

$$\dot{\vec{H}}_{/G} = \int \vec{r}_{/G} \times \vec{a}_{/G} dm$$

Shortcut (rigid objects) using moment of inertia

$$\dot{\vec{H}}_{/G} = \frac{d}{dt} (\vec{H}_{/G})$$

$$\hookrightarrow \dot{\vec{H}}_{/G} = \underline{\underline{I}} \cdot \vec{\omega}$$

$$\left[\begin{array}{l} \text{Aside: } \vec{\omega} = \vec{\omega}_1 + \vec{\omega}_2, \quad \boxed{\vec{\omega}_{P/Q} = \vec{\omega}_{P/I} + \vec{\omega}_{I/Q}} \\ \text{- why? Because small rotations commute} \end{array} \right]$$

$$\underline{\underline{I}} = I_3 \hat{e}^B \hat{e}^B + I_1 \underbrace{(\hat{e}^A \hat{e}^A + \hat{e}^B \hat{e}^B + \hat{k}^B \hat{k}^B)}_{\underline{\underline{1}}_1} - I_1 \hat{k}^B \hat{k}^B$$

$$= (I_3 - I_1) \hat{k}^B \hat{k}^B + I_1 \underline{\underline{1}}_1$$

$$\vec{H}_{/G} = \underline{\underline{I}} \cdot \vec{\omega}$$

$$= [(I_3 - I_1) \hat{k}^B \hat{k}^B + \underline{\underline{1}}_1 I_1] \cdot (\omega_p \hat{k} + \omega_s \hat{k}')$$

$$= \omega_p (I_3 - I_1) \hat{k}^B \underbrace{(\hat{k}^B \cdot \hat{k})}_{\cos \phi} + \omega_s (I_3 - I_1) \hat{k}^B \underbrace{\hat{k}^{B'} \cdot \hat{k}^B}_{\underline{\underline{1}}_1} \hat{k}^B$$

$$I \omega_p \hat{k} + \omega_s I_1 \hat{k}^B$$

$$\vec{H}_{/b} = (I_1 \omega_P) \hat{k} + (\omega_P (I_3 - I_1) \cos \phi + \omega_s I_3) \hat{k}^P$$